CISS245: Advanced Programming
Quiz q6001

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Q1. The class `Date` has member functions `add_y(int inc_y)`, `add_m(int inc_m)`, and `add_d(int inc_d)` that respectively increments the year, month, and day of a `Date` object. Complete the following function that increments the year, month, day of a Date object.

Answer:

```
void Date::add_y_m_d(int inc_y, int inc_m, int inc_d)
{
  add_y(inc_y);
  add_m(inc_m);
  add_d(inc_d);
}
```

Q2. You are given

```
class Position
{
private:
    int x_, y_;
};
```

Add enough code to the above so that the following works:

```
Position p;
p.set_x(0); // p.x_ is set to 0
p.set_y(1); // p.y_ is set to 1
p.print();  // print "<0, 1>" (without double quotes)
```

Answer:

```
class Position
{
public:
void set_x(int x)
{
  x_ = x;
}

void set_y(int y)
{
  y_ = y;
}
```

```
void print() const
{
  std::cout << '<' << x_ << ", " << y_ << '>' << std::endl;
}

private:
    int x_, y_;
};
```

Q3. You are given

```
class Array
{
public:
    void push_back(int v)
    {
        if (size_ < capacity_)
        {
            x_[size_] = v;
            ++size_;
        }
    }
private:
    int x_[1024];
    int size_;
    int capacity_;
};
```

Add a method `is_ascending()` that will return true of `x_[0]`, ..., `x_[size_ - 1]` is in ascending order. For instance

```
Array a;
a.push_back(1);
a.push_back(3);
a.push_back(3);
a.push_back(5);
std::cout << a.is_ascending() << '\n'; // prints 1
```

Answer:

```
class Array
{
public:
    void push_back(int v)
    {
        if (size_ < capacity_)
        {
            x_[size_] = v;
            ++size_;
        }
```

```
    }

    // YOUR ANSWER HERE
    is_ascending()
    {
      for (int i = 0; i < size_ - 1; ++i)
      {
        if (x_[i] > x_[i + 1])
        return false;
      }
      return true;
    }

private:
    int x_[1024];
    int size_;
    int capacity_;
};
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.